PG No. 13 C_{4v} $4mm$ [tetragonal]

* character table

C_{4v}	$1(1)$	$2_{001}(1)$	$4^{+}_{001}(2)$	$\mathrm{m}_{100}(2)$	$\mathrm{m}_{110}(2)$
A_1	1	1	1	1	1
A_2	1	1	1	−1	−1
B_1	1	1	−1	1	−1
B_2	1	1	−1	−1	1
E	2	−2	0	0	0

* polar ↔ axial conversion

A_1 (A_2)　B_1 (B_2)　E (E)　A_2 (A_1)　B_2 (B_1)

* symmetric product

	A_1	A_2	B_1	B_2	E
A_1	A_1	A_2	B_1	B_2	E
A_2		A_1	B_2	B_1	E
B_1			A_1	A_2	E
B_2				A_1	E
E					$A_1 + B_1 + B_2$

* anti-symmetric product

A_1	A_2	B_1	B_2	E
−	−	−	−	A_2